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CHECKLIST

SMALL CRAFT - Lithium-ion batteries

Not yet harmonised

Ref.: ISO 23625:2025 (ISO 23625:2025)

Checklist_Evaluation_Module B_G en260703

Watercraft manufacturer:	
Watercraft model name:	

Subject to check **Clause** **Requirements** **Checked ?**

This checklist is based on ISO 23625:2025 which has following scope:
Specifing requirements and recommendations for the selection and installation of lithium-ion batteries for boats, as well as requirements for the safety information provided by the manufacturer.
ISO 23625:2025 is applicable to lithium-ion batteries and battery systems with a capacity greater than 500 Wh used on small craft for providing power for general electrical loads and/or to electric propulsion systems.
It is primarily intended for manufacturers and battery installers.

Information about Li-Ion battery checklist:
1. For definitions, see end of checklist which is outside of print envelope.
2. If the watercraft has several li-ion battery banks on board (e.g. for household battery, bow thruster, electrical propulsion), please fill the below information for each one separetly.
3. This checklist provides an overview but does not replace the need for more in detail documentation such as technical datasheet(s), wire diagram, safety philophisy for the Energy Storage Space (ESS), etc.

Number of battery banks on board:

Section to be filled in for each battery bank

Battery bank details:

Battery bank consists of follwing batteries (name of the battery manufacturer and exact type of batteries):

Battery bank consists of following number of batteries:

Battery meets the requirements of IEC 62619 and IEC 62620?

Battery bank electrical connected parallel, series or both?

Watercraft manufacturer:	
Watercraft model name:	

If series and paralleling installations are done, they comply with the lithium-ion battery manufacturer's specifications?

If series and paralleling installations are done and the battery manufacturer specifies a voltage deviation limit between individual battery voltages, this is met before the batteries are paralleled or connected in series to prevent excessive battery to battery current?

When connection in parallel, a means is provided to prevent overcurrent conditions?

Batteries of one battery bank have identical chemistry?

Battery bank voltage (V):

Battery bank capacity (in Ah or Wh):

Battery bank location:

Battery bank used for propulsion:

Watercraft manufacturer:	
Watercraft model name:	

If battery bank greater than 1500 WH is used for propulsion, state of health provided for the battery system?

Battery banknot used for propulsion serves following electrical consumers:

Lithium-ion battery systems shall be installed in accordance with the battery manufacturer’s recommendations.	
The battery system is sized in accordance with the application and the battery manufacturer’s defined operating limits, and with the appropriate C rating listed in the battery specifications.	
Lithium-ion batteries and their systems are be installed in accordance with the requirements of ISO 13297. NOTE Lithium-ion batteries are not subject to routine electrolyte leakage or routine release of gas, therefore requirements regarding electrolyte containment and routine gas ventilation cannot apply to lithium-ion battery installations.	
Battery Management System (BMS) installed to controll the batteries and maintain the battery manufacturer's specified Safe Operation Limit (SOL)?	
Battery Management System (BMS) fully integrated in batteries or part thereof externally installed?	
Battery Management System (BMS) provides a battery cut-off if hazardous conditions for all batteries? Note 1 A BMS can be external or internal to the battery. Note 2 In the event of an individual BMS that disconnects, battery bank capacity can be reduced with no notification to the vessel operator.	
The BMS shall be designed and tested to manage the following: a) safety-related elements: 1) overcharging, to protect the lithium-ion battery from excessive charging; 2) overdischarging, to protect the lithium-ion battery from excessive discharging; 3) over and under temperature, to protect the lithium-ion battery from excessive temperature extremes; b) performance-related elements: balancing, to provide for automatic balancing of cells or strings of cells.	

Watercraft manufacturer:	
Watercraft model name:	

If a shut-down condition is approaching for propulsion systems or other critical system (equipment vital to safety), a BMS or system notifies the operator with a visual and/or audible alarm, clearly perceptible from the main helm position, prior to disconnecting the battery from the DC system.	
Battery Disconnect Device: integrated or external?	
Multiple contactors are permitted (HVC, LVC, plus main), each providing specific protection from high voltage, low voltage and load isolation. A single contactor is permitted, if the control system provides for protection from all conditions.	
Safe Operation Limit (SOL) by battery supplier is defined for: - high and low voltage limits - charging and discharging limits - temperature limits - installation specifications.	
Installation takes the suppliers specification into account for temperatures? This extends to areas that can be heated from the sun or other heat sources.	
The system connections and BMS electronics are protected from corrosion?	
Installation takes the suppliers specification for shock and vibrations into account?	
Installation takes the suppliers specification of the IP rating into account with regard to locations with flooding or momentary submersion?	
Installation takes the suppliers specification with regard to ignition protection according ISO 8846 into account?	
For craft in long-term storage, the battery installation shall follow the manufacturer’s recommended battery storage procedures, based on ambient temperature, connected charging sources and parasitic loads.	
No electrical connections are made directly to the li-ion battery, so that it would bypass the BMS?	
The battery installation have an overcurrent protection that does not exceed the current interrupting capacity (AIC-rating) of the BMS?	
Appropriate measures for balancing and overcurrent protection between batteries are taken?	
Output control from charging sources is within the battery manufacturer’s specified ranges?	
Charging sources shall be operated/controlled to meet the charging profile recommendations provided by the lithium-ion battery or cell manufacturer.	
If a battery manufacturer specifies a voltage deviation limit between individual battery voltages, it shall be met before the batteries are paralleled or connected in series to prevent excessive battery to battery current.	
Fire protection: the battery manufacturer's recommendation(s) are followed to protect against fire hazards, including venting and specific fire suppression mechanisms (e.g. specific fire-extinguishing medium).	
A documented safety philosophy is provided for the Energy Storage Space (ESS) how the safety of passengers, crew and vessel is ensured.	

compliant: Yes or √ not applicable: NA follow up on variation report: Rpt or X

Watercraft manufacturer:	
Watercraft model name:	

The ESS safety philosophy takes into account: gas development hazard (toxic, flammable, corrosive); - fire hazard; - necessary detection, monitoring and alarm systems (off-gas detection, fire detection, etc.) and ventilation; - explosion hazard; - ventilation handling in case of off-gas release and/or fire; - external hazards (fire, water ingress, etc.).	
In case of a cell venting, the battery installation allows for cell venting such that the safety of persons onboard is not endangered.	

Battery Manufacturer's safety information and operational's manual

Subject to check	Requirements	Checked ?
Cell chemistry and cell design	[Yes ?]	
Capacity ratings including discharge rate or time, and temperature	[Yes ?]	
Description of the primary features of the battery with regard to safety (e.g. use of touch-safe covers, operation of the BMS, cell venting, manual service disconnects)	[Yes ?]	
Reset procedures for the battery system is a shutdown event occurs	[Yes ?]	
Detailed description of operating requirements necessary to ensure safe operation, with details of the SOL, including high and low voltage limits, relevant charge and discharge characteristics and temperature limits, and any external relays and other features necessary to ensure operation within the SOL;	[Yes ?]	
Requirements, if any, for the control of charging sources	[Yes ?]	
Effects of an external fire or excessive external heating, together with a description of measures to be taken, if any, to guard against external hazards such as fire and impact, including any venting and specific fire suppression mechanisms (e.g. specific fire-extinguishing medium);	[Yes ?]	
Information about the toxic, flammable or explosive gases that can be released given a thermal runaway event;	[Yes ?]	
Description on the limits to connecting batteries in series and parallel, and a description of a safe means of making the connections up to the specified limits, including overcurrent and other required protective mechanisms;	[Yes ?]	
Description of the hazards, if any, that occur with submersion in freshwater and saltwater;	[Yes ?]	
Battery short circuit current or the rating of the overcurrent protection device to protect the circuit;	[Yes ?]	
Requirements for restraining the battery in the craft, including information about shock and vibration and movement limits of the battery;	[Yes ?]	
Storage and winterization procedures provided	[Yes ?]	
Limitations on the working angle of the battery system provided, if any;	[Yes ?]	
Information regarding the recycling and/or disposal of lithium-ion batteries;	[Yes ?]	
Caution about combining different battery types.	[Yes ?]	

compliant: Yes or √ not applicable: NA follow up on variation report: Rpt or X